

Haiyang Liu, PhD

Aquatic Animal Genomics and Molecular Breeding

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I received my PhD in Genetics from the Institute of Hydrobiology, Chinese Academy of Sciences (University of Chinese Academy of Sciences) in 2017 and joined the Pearl River Fisheries Research Institute, Chinese Academy of Fishery Sciences in the same year, where I have worked since. My research covers the development and utilisation of aquatic animal germplasm resources, genetic dissection of economically important traits, genomic selection breeding, and the application of machine learning in aquaculture breeding. I have established an end-to-end technical system spanning genome assembly and annotation, population-genetic analysis, genome-wide association studies (GWAS) and genomic selection. I have completed chromosome-level genome assemblies for more than ten aquatic species, including blotched snakehead, small snakehead, mud carp, darkbarbel catfish and *Siniperca obscura*; carried out a systematic survey of snakehead germplasm resources, characterising the genetic diversity, population structure and demographic history of the main cultured populations; dissected the genetic basis of growth and sex determination in snakeheads and other fishes, developing multiple sex-specific molecular markers; led the development of 50K liquid SNP breeding arrays for northern snakehead and blotched snakehead, built the associated genomic selection system, and explored machine-learning algorithms for accurate germplasm identification, complex-trait prediction and genomic estimated breeding values; and contributed to the breeding of the all-male hybrid snakehead variety “Xiongli No. 1” and its demonstration and extension in major production areas — forming a complete chain from genome analysis, germplasm assessment, trait dissection and array development through intelligent prediction and breeding application to commercialisation. To date I have published 31 SCI papers as first or corresponding author (79 SCI papers in total), with over 1,200 citations and an H-index of 20; I have led 12 research and commercialisation projects (total funding CNY 2.58 million) and participated in 6 national and provincial projects; I hold 12 granted patents (5 as first inventor, 2 of which have been commercially transferred); I contributed to the nationally approved hybrid snakehead variety “Xiongli No. 1” (GS-02-003-2022); I have received 7 science and technology awards, including the Fan Li Science and Technology Award (Second Prize, 2025); I participated in drafting 6 industry standards and 1 local standard; and I serve as a guest editor for *Animals* and *Fishes* and a reviewer

for international journals including *Nature Communications*, *Scientific Data* and *iScience*.

31

First/corresponding-author SCI papers (79 in total)

1200+

Citations • H-index 20

12

Granted patents (5 as first inventor)

7

Industry & local standards

12

Funded projects as PI

1

Nationally approved new variety

Research areas

Genome resources and annotation

Chromosome-level and high-quality genome assemblies (PacBio HiFi / ONT / Hi-C) for snakeheads, mud carp, bagrid catfishes and siniperine fishes, with gene-structure annotation from multi-species protein and RNA evidence, repeat and functional annotation. Completeness is tracked with BUSCO and compleasm; gaps are diagnosed and closed by targeted rescue.

Sex determination and trait genetics

High-density linkage mapping, QTL fine mapping, GWAS and transcriptome integration to dissect sex determination regions and growth-related loci; sex-specific molecular markers developed for snakeheads (*C. argus*, *C. maculata*, *C. striata*, *C. asiatica*), mud carp, darkbarbel catfish and bighead/silver carp, supporting all-male and monosex fry production.

Molecular breeding and SNP arrays

Led development of 50K liquid SNP breeding arrays for northern snakehead and blotched snakehead (patented and commercially licensed), including locus selection, probe design, genotyping quality control and cross-population validation; application to genomic selection, population assignment and parentage assignment.

Population genetics and germplasm assessment

Resequencing- and reduced-representation-based assessment of genetic diversity, population structure, linkage disequilibrium, selection signatures and demographic history in wild and selected populations, informing germplasm conservation and parental selection.

Machine learning and intelligent breeding

Applying machine learning across the breeding pipeline: accurate germplasm identification and population assignment from genomic and phenotypic data, prediction of complex traits, and genomic estimated breeding value (GEBV) assessment — supporting low-density-array genomic selection and parental selection.

Education and experience

2007.09–2011.06	Southwest University — BSc in Aquaculture (Bachelor of Agriculture)
2011.09–2017.06	Institute of Hydrobiology, Chinese Academy of Sciences (University of Chinese Academy of Sciences) — MSc–PhD in Genetics (PhD)
2017.07–present	Pearl River Fisheries Research Institute, CAFS — Dept. of Aquatic Germplasm Resources and Genetic Breeding, Assistant Researcher

First/corresponding-author papers (31)

- Liu HY**#, Xie W#, Liu JF, Cui TX, Luo Q, Fei SZ, Zhu XP, Chen KC, Ou M*, Zhao J*. Chromosome-level genome assembly of blotched snakehead (*Channa maculata*) using PacBio HiFi and Hi-C sequencing. **Scientific Data**, 2026, Article in Press. [First author](#) [DOI](#)
- Liang YY, Liu HJ, Cui K, Li J, Fang T, Gao N, Chen C, Zhao XX, Yang K, Wang X, Zhao J, Lu WX*, **Liu HY***. Chromosome-level genome assembly of *Coreoperca whiteheadi*. **Scientific Data**, 2026, Article in Press. [Corresponding author](#) [DOI](#)
- Liu HY***, Liu HJ, Cui K, Lu WX, Li J, Fang T, Gao N, Chen C, Zhao XX, Yang K, Lin YF*, Liang YY*. Chromosome-level genome assembly of the *Siniperca obscura*. **Scientific Data**, 2026, 13: 333. [Corresponding author](#) [DOI](#)
- Liu HY**, Li XY, Xie W, Wang Y, Ou M, Luo Q, Fei SZ, Zhang XC, Chen WF, Zhu XP, Chen KC, Zhao J*. A genome-wide association study identifies candidate SNPs and genes associated with body weight in the northern snakehead (*Channa argus*) using a 50K SNP array. **Animals**, 2026, 16: 2471. [First author](#) [DOI](#)
- Zhang J, Cui TX, Liu JJ, Liu JF, Xia WW, Ou M, Luo Q, Fei SZ, Zhu XP, Chen KC, Zhao J*, **Liu HY***. Development of a 50K liquid SNP array and its application in GWAS and population genomics of northern snakehead (*Channa argus*). **Aquaculture**, 2026, 743737. [Corresponding author](#) [DOI](#)
- Zhang J, Cui TX, Yang ZW, Liu JJ, Liu JF, Ou M, Luo Q, Fei SZ, Zhu XP, Chen KC, Zhao J*, **Liu HY***. Population identification in northern snakehead (*Channa argus*) through machine learning integration of informative SNPs from whole-genome sequencing and 50K SNP array. **Aquaculture**, 2026, 743659. [Corresponding author](#) [DOI](#)
- Xie W#, Xia WW#, Liu JJ, Liu JF, Wang Y, Ou M, Luo Q, Fei SZ, Chen KC, Jiang H, Zhao J*, **Liu HY***. Development of a multiplex microsatellite genotyping system and its application in parentage assignment of hybrid snakehead (*Channa argus* ♀ × *C. maculata* ♂). **Aquaculture Reports**, 2026, 49: 103691. [Corresponding author](#) [DOI](#)
- Zhu CK*, Zhu JY, Wang Y, Lu C, Ouyang ZS, Pan ZJ, Li SC, Sun YH, Chang GL, Zhao HT, Zhang L*, **Liu HY***. Identification of growth-related SNPs and genes through genome-wide association studies in *Pseudobagrus ussuriensis*. **Marine Biotechnology**, 2026, 28(3): 107. [Corresponding author](#) [DOI](#)
- Gan BJ, Wei LJ, Ma YX, Mo FL, Xiao S, Lu YD, Liu K, Yang BL, Zhang S*, **Liu HY***. A chromosome-level reference genome assembly of the giant pangasius (*Pangasius sanitwongsei*). **Scientific Data**, 2026, 13: 130. [Corresponding author](#) [DOI](#)
- Liu HY**#, Zhang J#, Cui TX, Zhang XC, Li KB, Wang F, Luo Q, Fei SZ, Chen BX, Zhu CK, Chen KC, Zhu XP, Li BJ, Zhao J*, Fang LZ*, Ou M*. A chromosome-level genome assembly of the male darkbarbel catfish (*Pelteobagrus vachelli*) using PacBio HiFi and Hi-C data. **Scientific Data**, 2025, 12(1): 351. [First author](#) [DOI](#)
- Yu LY#, **Liu HY**#, Wei J, Zhang WC, Hong KH, Li HR, Zhou QY, Su QY, Liufu B, Zhu XP*, Wang YK*. A high-quality chromosome-level genome assembly and annotation of the giant freshwater prawn (*Macrobrachium*

- rosenbergii). **Scientific Data**, 2025, 12(1): 1181. [Co-first author](#) [DOI](#)
- 12 Liu HY#, Liu JF#, Cui TX, Zhang J, Luo Q, Fei SZ, Chen KC, Zhu XP, Ou M*, Zhao J*. A chromosome-level reference genome assembly of the small snakehead (*Channa asiatica*). **Scientific Data**, 2025, 12(1): 1159. [First author](#) [DOI](#)
- 13 Liu HY*, Yao JJ, Dong MD, Yi MS, Liu W*. Chromosome-level genome assembly of the marble goby (*Oxyeleotris marmorata*). **Scientific Data**, 2025, 12(1): 1596. [Corresponding author](#) [DOI](#)
- 14 Liu HY#, Cui TX#, Zhang J, Zhang Y, Luo Q, Fei SZ, Chen KC, Zhao J*, Ou M*. Identification of sex determination region and candidate genes in blotched snakehead (*Channa maculata*) through genome-wide association studies and transcriptome analysis. **Aquaculture Reports**, 2025, 42: 102825. [First author](#) [DOI](#)
- 15 Cui TX#, Liu HY#, Zhang J, Luo Q, Fei SZ, Chen KC, Zhao J*, Ou M*. Development and evaluation of a 50K SNP array for blotched snakehead (*Channa maculata*). **Aquaculture**, 2025, 598: 742020. [Co-first author](#) [DOI](#)
- 16 Cui TX, Zhang J, Ou M, Luo Q, Fei SZ, Chen KC, Zhao J*, Liu HY*. Potential of genome-wide association studies to improve genomic selection for growth traits in blotched snakehead (*Channa maculata*). **Aquaculture**, 2025, 596: 741895. [Corresponding author](#) [DOI](#)
- 17 Zhang J, Cui TX, Ou M, Luo Q, Fei SZ, Zhang XC, Li KB, Wang F, Chen BX, Zhu XP, Chen KC, Zhao J*, Liu HY*. GWAS and transcriptome analysis reveal male heterogametic sex-specific markers and candidate genes in darkbarbel catfish (*Pelteobagrus vachelli*). **Aquaculture**, 2025, 743312. [Corresponding author](#) [DOI](#)
- 18 Liu HY#, Cui TX#, Liu HJ, Zhang J, Luo Q, Fei SZ, Chen KC, Zhu XP, Zhu CK, Li BJ, Fang LZ, Zhao J*, Ou M*. Chromosome-level genome assembly of the mud carp (*Cirrhinus molitorella*) using PacBio HiFi and Hi-C sequencing. **Scientific Data**, 2024, 11(1): 1249. [First author](#) [DOI](#)
- 19 Liang YY, Liu HJ, Lu WX, Li J, Fang T, Gao N, Chen C, Zhao XX, Yang K, Liu HY*. Chromosome-level genome assembly of the smallscale yellowfin (*Plagiognathops microlepis*). **Scientific Data**, 2024, 11(1): 1234. [Corresponding author](#) [DOI](#)
- 20 Liu HY#, Zhang J#, Cui TX, Xia WW, Luo Q, Fei SZ, Zhu XP, Chen KC, Zhao J*, Ou M*. Genome-wide association studies (GWAS) and transcriptome analysis reveal male heterogametic sex-determining regions and candidate genes in northern snakeheads (*Channa argus*). **International Journal of Molecular Sciences**, 2024, 25(20): 10889. [First author](#) [DOI](#)
- 21 Liu HY#, Xia WW#, Ou M, Luo Q, Zhang XC, Fei SZ, Huang SJ, Zhao XJ, Zhang J, Cui TX, Xiong BY, Wu GH, Chen KC, Zhao J*. A genome-wide association study to identify growth-related SNPs and genes in blotched snakehead (*Channa maculata*). **Aquaculture Reports**, 2024, 35: 101932. [First author](#) [DOI](#)
- 22 Liu HY#, Xia WW#, Li BJ, Liu L, Wang YK, Luo Q, Ou M, Zhu XP, Chen KC, Zhao J*. Sex-specific markers developed by 2b-RAD and genome sequencing reveal an XX/XY sex-determination system in mud carp (*Cirrhinus molitorella*). **Aquaculture**, 2023, 565: 739131. [First author](#) [DOI](#)
- 23 Liu L, Liu HY*, Xia WW, Chen KC, Luo Q, Ou M, Zhao J*. Isolation and characterization of 66 SNP markers in blotched snakehead (*Channa maculata*) using 2b-RAD sequencing. **Conservation Genetics Resources**, 2023, 15(4): 215-220. [Corresponding author](#) [DOI](#)
- 24 Zhu CK*, Liu HY*, Pan ZJ, Cheng L*, Sun YH, Wang H, Chang GL, Wu N, Ding HY, Zhao HT, Zhang L*, Yu XS. Insights into chromosomal evolution and sex determination of *Pseudobagrus ussuriensis* (Bagridae, Siluriformes) based on a chromosome-level genome. **DNA Research**, 2022, 29(4): dsac028. [Corresponding author](#) [DOI](#)
- 25 Liu HY, Luo Q, Ou M, Zhu XP, Zhao J*, Chen KC*. Construction of a high-density linkage map and QTL detection of growth and sex in blotched snakehead (*Channa maculata*). **Aquaculture**, 2021, 736541. [First author](#) [DOI](#)

- 26 Zhu CK*, **Liu HY***, Cheng L, Pan ZJ, Chang GL, Wu N, Wang H, Ding HY, Fang Y, Wang L. Identification of sex-specific sequences through 2b-RAD sequencing in *Pseudobagrus ussuriensis*. **Aquaculture**, 2021, 539: 736639. [Corresponding author](#) [DOI](#)
- 27 **Liu HY**, Luo Q, Ou M, Zhu XP, Zhao J*, Chen KC*. High-density genetic linkage map and QTL fine mapping of growth and sex in snakehead (*Channa argus*). **Aquaculture**, 2020, 519: 734760. [First author](#) [DOI](#)
- 28 Zhou Y#, **Liu HY#**, Wang XH, Fu BD, Yu XM, Tong JG*. QTL fine mapping for sex determination region in bighead carp (*Hypophthalmichthys nobilis*) and comparison with silver carp (*Hypophthalmichthys molitrix*). **Marine Biotechnology**, 2020, 22: 41-53. [Co-first author](#) [DOI](#)
- 29 **Liu HY**, Pang MX, Yu XM, Zhou Y, Fu BD*, Tong JG. Sex-specific markers developed by next-generation sequencing confirmed an XX/XY sex determination system in bighead carp (*Hypophthalmichthys nobilis*) and silver carp (*Hypophthalmichthys molitrix*). **DNA Research**, 2018, 25: 257-264. [First author](#) [DOI](#)
- 30 **Liu HY**, Fu BD, Pang MX, Feng X, Yu XM, Tong JG*. A high-density genetic linkage map and QTL fine mapping for body weight in crucian carp (*Carassius auratus*) using 2b-RAD sequencing. **G3: Genes, Genomes, Genetics**, 2017, 7: 2473-2487. [First author](#) [DOI](#)
- 31 **Liu HY**, Fu BD, Pang MX, Feng X, Wang XH, Yu XM, Tong JG*. QTL fine mapping and identification of candidate genes for growth-related traits in bighead carp (*Hypophthalmichthys nobilis*). **Aquaculture**, 2016, 465: 134-143. [First author](#) [DOI](#)

Research projects

As principal investigator (12 in total)

- Guangdong Provincial Department of Science and Technology, General Programme: Construction and optimisation of genomic selection for blotched snakehead using low-density SNP breeding arrays (2026A1515011962, 2026–2028, CNY 100,000)
- Hainan Provincial Department of Science and Technology, General Programme: Fine mapping and functional analysis of the sex determination region in mud carp (326MS0347, 2026–2028, CNY 80,000)
- Guangdong "Hundred-Thousand-Ten Thousand Project" rural science and technology special commissioner: Demonstration and extension of breeding and healthy aquaculture technologies (KTP20240444, 2024–2027, CNY 200,000)
- CAFS Central Public-interest Scientific Institution Basal Research Fund: AI-driven genomic selection using low-density SNP arrays (2026SJHX1, 2026, CNY 90,000)
- CAFS Central Public-interest Scientific Institution Basal Research Fund: AI-driven genomic selection breeding technologies (2025SJHX1, 2025, CNY 120,000)
- Patent commercialisation: SNP array for northern snakehead and its application (Foshan Gangzhong Fishery Development Co., Ltd., 2025–2028, CNY 750,000)
- Patent commercialisation: sex-determination marker for small snakehead and its application (Guangdong Bairong Aquatic Seed Group, CNY 300,000)
- Contract research: genomic selection services for snakehead (Qingyuan Zhuoyue Aquatic Co., CNY 240,000)

- Contract research: breeding services for albino catfish (Neijiang Xinyongling Construction & Development Co., CNY 400,000)
- Guangzhou Municipal Science and Technology Bureau, Basic and Applied Basic Research (young PhD): Characterisation of the Y chromosome in northern snakehead and its application in all-male production (202102021070, 2021–2023, CNY 50,000, completed)
- CAFS Central Public-interest Scientific Institution Basal Research Fund: Effects of EM (effective microorganisms) on healthy aquaculture of all-male hybrid snakehead (2021SJ-XT2, 2021–2022, CNY 100,000, completed)
- CAFS Central Public-interest Scientific Institution Basal Research Fund: Fine mapping of growth- and sex-related QTL and candidate gene screening in blotched snakehead (2019KQ-001, 2019–2020, CNY 150,000, completed)

As participant (national / provincial)

- NSFC General Programme: Mapping and functional identification of the sex determination gene in blotched snakehead (32373127, 2024–2027, total budget CNY 500,000)
- China Agriculture Research System for Freshwater Fish: Germplasm resources and genetic improvement of snakeheads (CARS-46, 2024–2027, total budget CNY 3,000,000)
- National Key R&D Programme "Blue Granary": Genomic selection and monosex control technologies for important cultured fishes (2018YFD0901201, 2018–2022, total budget CNY 9,220,000, completed)
- Guangdong Provincial Seed Industry Revitalisation Programme: Breeding and extension of hybrid snakehead "Xiongli No. 2" (2022-SPY-00-016, total budget CNY 3,000,000)
- Government procurement project: Purification and rejuvenation of largemouth bass and northern snakehead (JJ-2020-009, 2021–2025, total budget CNY 5,000,000)
- CAFS Science and Technology Innovation Team: Snakehead and grass carp genetic breeding innovation team (2020TD34, total budget CNY 2,490,000)

Awards, new variety and standards

Awards

- Breeding and industrial application of improved hybrid snakehead — Fan Li Science and Technology Award, Science and Technology Progress Second Prize, China Society of Fisheries (2025)
- Key technologies for breeding and industrialisation of hybrid snakehead — CAFS Science and Technology Award (2025); Lin Haoran Animal Science Award, First Prize (2024); Lingnan Animal and Plant Science and Technology Award, First Prize (2024)
- Creation and extension of new snakehead varieties — Zhong Lin Aquatic Seed Industry Science and Technology Award (2022)
- Extension award for "Wuban" hybrid snakehead fry production and healthy aquaculture (Department of Agriculture and Rural Affairs of Guangdong, 2020)

New variety

- Hybrid snakehead "Xiongli No. 1" (GS-02-003-2022) — selected among the 2023 outstanding achievements in new fishery technologies, products and equipment

Standards (7: 6 industry, 1 local)

- Mud carp broodstock and fry (SC/T 1185-2025) | Small snakehead broodstock and fry (SC/T 1186-2025)
- Technical specification for artificial breeding of northern snakehead (SC/T 1174-2023) | *Siniperca scherzeri* (SC/T 1171-2023)
- Blotched snakehead broodstock and fry (SC/T 1162-2022) | *Scatophagus argus* (SC/T 1152-2021)
- Guangdong local standard: hybrid snakehead "Xiongli No. 1" (DB44/T 2385-2022)

Patents

- 12 granted patents (11 invention, 1 utility model), 5 as first inventor: SNP array for northern snakehead (ZL 202411478644.1); sex marker for mud carp (ZL 202211273634.5); sex marker for small snakehead (ZL 202311194491.3); sex marker for striped snakehead (ZL 202211709489.0); sex marker for darkbarbel catfish (ZL 202311335874.8)
- Two first-inventor patents commercially licensed (SNP breeding array for northern snakehead; sex marker for small snakehead)

Academic service

- **Guest editor:** *Animals, Fishes*
- **Reviewer:** *Nature Communications, Scientific Data, Communications Biology, Zoological Research, iScience, Aquaculture, Aquaculture Reports, Marine Biotechnology, International Journal of Biological Macromolecules, Genomics, BMC Genomics, Comparative Biochemistry and Physiology Part D, Heliyon, Genes, Biology, Animals, Fishes*
- **Supervision:** 5 graduate students completed their degrees
- **Postgraduate supervisor:** Hainan University; Huaiyin Normal University

Education and positions

Contact

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